Gaurav Shankar

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I am a Cranfield University Mechanical Engineer graduate with a deep passion for engineering and problem solving. I have a specialized focus on complex fluid flow analysis and computational aerodynamics. I am a committed and hands-on problem solver, confident in delivering impactful results in engineering roles. I am passionate about creating sustainable and efficient solutions to practical engineering challenges, creating new designs, computational analysis and understanding complex fluid flow analysis. I possess the ability to lead a team, manage people and resources, and build strong relationships with clients and internal stakeholders.

EDUCATION

MSc in Advanced Mechanical Engineering | Cranfield University, Cranfield, UK

10.2024 - 09.2025

Relevant Coursework: Structural Integrity, Fluid Mechanics and Loading, Computational Fluid Dynamics (CFD) for Renewable Energy, Engineering Stress Analysis, Engineering Design and Project Management, Assessing Risk and Failure.

- **Group Project:** Ammonia-Powered Hydrogen Drones. Contributed to a group project focused on the future of aerial mobility using ammonia-powered hydrogen drones.
- **Thesis Project:** Thesis Project: High-Temperature Corrosion behavior of Cold Rolled Additively Manufactured Stainless Steels. Researched and analyzed the high-temperature corrosion behavior of additively manufactured stainless steel 316L in molten solar salt **(Cranfield University has selected my master's thesis for inclusion in its archive in recognition of its quality, having been marked within the top 10% of my cohort's submissions or identified as an important work)**

BTech in Mechanical Engineering | Amrita Vishwa Vidyapeetam, Bengaluru, India

06.2019 - 09.2023

- **Relevant Coursework:** Mechanics of Solids, Finite Element Method (FEM), Computational Fluid Dynamics (CFD), Robotics, CAD/CAM, Kinematics of Machines, Automation and IoT, Linear Algebra.
- **Thesis Project:** Modular Serial-Chain Robotic Arm: Designed and developed a modular, plug-and-play serial-chain robotic arm to serve as an effective educational tool for teaching robotics principles.

WORK EXPERIENCE

Amrita Vishwa Vidyapeetham | Project Associate

Bengaluru, India | 10.2023 - 04.2024

- Conducted research and development for a funded project to develop a 2 and 6-axis robot module.
- Designed and programmed a robot using a Wave Share SC15 servo motor and an ESP32 microcontroller to demonstrate robotic movements and applications to students.
- Engineered new designs and modified existing robot architecture, resulting in a 20% weight reduction.
- Collaborated with one other team member to research and develop a new robot.

Airbus India Private Limited | In-Service Engineering Intern

Bengaluru, India | 03.2023 - 09.2023

- Automated and tested SRA (Smart Repair Assistant) tool, which boosted efficiency by 30% per job for in-service deliverables.
- Optimized team workflows and standardized damage mapping on aircraft, creating fixed repair modules that reduced solution time by 20 minutes per aircraft.

- Collaborated with a team of three design engineers specializing in aircraft airworthiness and structural design.
- We analyzed historical and current damage cases, gaining insights that led to faster repair solutions.
- Reduced the time to produce in-service aircraft documents by 100 minutes per document, setting new benchmarks for the team and enhancing the overall speed and safety of repair solutions.

MTRDC (DRDO) | Internship Trainee

Bengaluru, India | 06.2022 - 07.2022

- Conducted a comparative analysis and testing of a heat sink under natural and forced convection to prove the superior efficiency of forced convection.
- Simulated the thermal performance of both natural and forced convection methods using Flowtherm software, conducting 10 distinct test cycles to validate the results.
- Troubleshooted and resolved issues with Wire EDM equipment, minimizing downtime and restoring production within 4 hours, ensuring continuous operation of manufacturing processes.

Technical Skills CAD/SolidWorks, Catia, Abaqus, ANSYS-Fluent, ANSYS-Workbench, MATLAB/Simulink. Roboanalyzer, UNREAL engine, Fusion 360, Inventor Pro, Google Workplace, Python.

Soft Skills Strong oral communication. Collaborative approach. Confident and Curious mindset.

AWARDS & ACHIEVEMENTS

- Applied for patent titles "**Modular Serial-Chain Robot** "and the "**Patent ID 202341022970**" Filed on March 29,2023
- Awarded the best presented paper from AIR "Advances in Robotics" 2023 "**Modular Serial Chain Robot with Plug-and-Play Architecture for Effective Teaching and Learning**".
- Presented an international paper at IFToMM WC 2023 titled " **Virtual and Physical Components to Build Modular Robotic Arms for Effective Understanding of Denavit-Hartenberg (DH) Parameters** ".
- Two-time Winner of **AAROHAN (Amrita Annual Science Fest)** in 2022 and 2023 for the best robot design.
- Trained over **200+ students** on Drones and Robotics.

LANGUAGES & INTERESTS

- English- High Proficiency. German - A2 Level, Kannada – Native language.
- Reading, Chess, Cricket, Computer Game, Biking and Hiking, Aviation and Photography.

EXTRA- CURRICULAR

- Training conducted on Robotics and Kinematics for 50+students in August 2023.
- FORGE (Forum for Game Engineering) Vice president Jun 2021- Sep 2022
- JIDO (The Automation Club) President Jun 2021- Sep 2022
- Avishkara, Kalakriti and Kalanjali, the main organizer committee, managed more than a team of 25+ students to successfully organize the event.
- Volunteer for DAASTAN biggest fest in Amrita Vishwavidya Peetham in 2023.

REFERENCES

- Available upon request.